NEXUS

Better product decisions with connected evidence

Business value from industry ontology, agentic AI and product memory

For product leaders, engineering, manufacturing, quality and the supply chain



Illustrative industrial photograph. Mixabest / Wikimedia Commons, CC BY-SA 3.0. Not a Nexus customer installation.

The business need

A product decision can depend on an assembly revision, a requirement, a process and inspection evidence held by different teams. NEXUS helps people bring the relevant records into one review, preserve their meaning and understand what is still missing.

Value to the business

Less time reconstructing the product story. A clearer basis for review, reuse and accountable action.

Shared meaning makes connected evidence useful

Lifecycle continuity, knowledge reuse and accountable AI are becoming linked business priorities. A connected record helps only when the team understands what it represents and which version applies.

Business challenge	Consequence for the review	How NEXUS helps
Fragmented evidence	People search before they can evaluate.	Gather relevant records and recorded relationships.
Different meanings	Similar names can hide different concepts.	Use reviewed business definitions and mappings.
Unclear revision or scope	Earlier findings may no longer apply.	Keep source identity and the selected baseline visible.
AI without inspectable evidence	Advice becomes difficult to challenge.	Show evidence, limitations and the decision owner.

Industry ontology

An ontology defines concepts and their relationships. The embedded public PCA Industrial Data Ontology (IDO) snapshot supplies reusable industrial meaning. The ISO 23726-1 public reference describes the overall ontology-based interoperability approach. Embedding the reference does not establish ISO certification or customer alignment.

Application and customer meaning

NEXUS distinguishes physical parts, design definitions, documents and activities. Customer terminology connects through reviewed mappings. A requirement stays distinct from evidence that it is satisfied, and a similar part stays distinct from an approved substitute.



Value to the business

More consistent interpretation across teams without discarding the original source or local vocabulary.

Four capabilities that improve decision preparation

Capability	Business question	Outcome to assess
Decision support intelligence	What needs review if this component changes?	A clearer impact-review scope and visible gaps.
Similarity	Which existing parts merit investigation for reuse?	Less repeated search and explainable differences.
Recommendations	What should we investigate next, and why?	Evidence-backed options for specialist review.
360-degree view and Where Used	Where is it used, and which evidence supports it?	A shared product context across disciplines.

A practical change review

An engineer selects the intended component revision. NEXUS follows recorded assembly, requirement and process references. The team compares candidates using available properties and declared units, then reviews conflicts, missing evidence and suitability constraints. The owner decides what action is authorized.

What the 360-degree view includes

Product and BOM context, engineering and MBSE references, manufacturing activities, quality and QIF evidence, source versions and retained review outcomes. PLM, CAD and simulation references are useful when established source associations are available.

The decision boundary

A 360-degree view describes complementary perspectives, not complete coverage. Similarity is not interchangeability. Recorded traceability does not establish requirement satisfaction, BOM completeness or manufacturing readiness. Missing links remain questions.

Value to the business

Less fragmented investigation and a review pack that the responsible specialists can inspect.

Memory makes the next question more useful

Memory	Example in business work	Value
Short-term	After selecting an assembly: "What are its requirements?"	Avoid repeating the current successful conversation subject.
Long-term	"What did the earlier reviewed mapping outcome teach us?"	Reuse retained evidence and approved advisory lessons.
Contextual	"Does that finding apply to this product revision and task?"	Reduce misleading carryover between investigations.

Short-term continuity

Relevant recent successful turns refine follow-up retrieval within the same authorized scope. An independent question should not inherit an unrelated product. Failed or unsupported answers do not become evidence.

Long-term reuse

Retained conversations, workflow outputs and source references support later review. Approved revision-scoped mapping lessons can advise another mapping assessment after applicability checks. Earlier approval does not make a lesson universally valid.

Context that travels with the task

The selected item, ontology, revision, purpose and permitted access constrain recall and investigation. Changed evidence can require a fresh review. Optional memory integrations depend on configuration.

Value to the business

Less repeated explanation and investigation, with a clearer check that prior knowledge still applies. Memory is not automatic model training.

Services and workers carry the review through

Services provide inspectable business results

Business responsibility	Contribution	Value
Prepare trustworthy inputs	Import, schema-set, normalization and ontology work retain and interpret evidence.	A source record that can be reviewed.
Support the investigation	Graph and linked-resource work expose recorded relationships.	A connected basis for the business question.
Share reviewed outputs	Data preparation, product and catalog work support governed reuse.	A result with ownership and intended use.

Three worker responsibilities

Worker	Business work	Retained evidence
Data preparation	Executes authorized processing and analytics jobs.	Task status and prepared outputs.
Agent workflow	Carries out permitted investigation steps.	Step results, status and recovery checks.
Product delivery	Reconciles approved package delivery to the catalog.	Delivery outcome and registration evidence.

A background task can continue while the user moves to another review. After interruption, the retained status and outcome need inspection before repeating an unverified action. A worker restart does not authorize a duplicate publication.

Value to the business

Less manual coordination, with a visible distinction between work requested, work completed and results delivered.

Agents and task skills serve accountable people

Representative agent role	Contribution to the review
Intake and engineering parser	Inspect the supplied evidence and source structure.
Graph Analyst	Investigate where used, change context and comparisons.
Ontology Governor	Coordinate mapping/merge proposals and permitted publication requests.
Ontology Steward	Assess meaning, ambiguity and retained validation evidence.
Data Product Governor	Coordinate authorized product publication and retained outcomes.
Data Quality Monitor	Inspect quality and processing evidence.

Skills are defined business actions

Retrieve evidence. Trace recorded dependencies. Compare options. Validate within stated coverage. Explain findings in business vocabulary. Prepare a retained proposal. Here, skills means registered permitted task capabilities, not an independent skills marketplace.

Personas and the decisions they own

Product leaders set priorities and pilot outcomes. Design engineers assess reuse and constraints. Systems engineers review requirement/model traceability. Manufacturing planners assess BOM and process context. Quality specialists review characteristic and inspection evidence. Stewards and approvers resolve meaning and authorize the intended scope.

Value to the business

Clearer responsibility at each handoff. Agents prepare and assess evidence; authorized people own the business decision.

One review approach can serve different industries

Industry	Illustrative decision	Outcome to evaluate
HiTech	Review a component change across assemblies and requirements.	Time to prepare a reuse or impact review.
Oil and gas	Review equipment specifications and operating context.	Effort to assemble related engineering evidence.
Food and beverages	Review product, process and quality references.	Visibility of missing specification associations.
Pharma	Review material, process and test evidence by version.	Effort to reconstruct a specialist evidence trail.

These are customer-pilot scenarios. Each needs supplied domain records, reviewed mappings and acceptance criteria. They are not prevalidated industry solutions or claims of food-safety, GxP, regulatory or batch-release certification.

Proposed, approved and released are different states

A proposal retains the evidence and intended scope. Review assesses findings and open questions. Authorization permits the specified operation. A verified outcome and receipt show what completed. Central Change Control provides visibility and links to the responsible workflow.

Changed information prompts review

Supported baselines and Tracked Resource Set (TRS) notifications can identify source changes and refresh needs. They support scoped review rather than silently authorizing another release. NEXUS does not replace a complete enterprise change-order or global configuration-management process.

Value to the business

A clearer chain of accountability when product information or review assumptions change.

A focused pilot establishes value with evidence

Reporting that teams can reuse

Choose registered business records and an agreed baseline. Define dimensions such as product, classification and revision. Agree formulas, units, denominators and effectivity before interpreting measures. Review the prepared analysis and intended use, then publish an owned result for discovery and reuse.

There is no universal automatic KPI library. Counts do not prove completeness, suitability or satisfaction. The business owner agrees what the measures mean.

Pilot measure	Definition to agree before the pilot
Review preparation effort	Elapsed effort to assemble the agreed evidence pack for equivalent scope.
Evidence coverage	Sampled records with identifiable source and revision context, with an agreed denominator.
Open questions	Unresolved relationships and elapsed time to disposition in the selected scope.
Reuse preparation	Effort to compare candidates or prepare an analysis another team can inspect.

Value to the business

Choose one product, one important question and one accountable owner. Compare the pilot with the current process before expanding. No generic savings percentage or guaranteed return is assumed.

Industry references

PCA: Ontology-based Interoperability for Industrial Data, ISO 23726-1 public reference

NIST: Digital Thread for Manufacturing

DMSC: Quality Information Framework

NIST: Identity and Authority of Software Agents, February 2026

AI-generated planning or narratives require an available configured model. Connected source coverage and customer acceptance remain prerequisites.